

1. (a) Explain the notion of unbiasedness and efficiency by using the sample average $\bar{Y}$ as an example.
(b) Explain the Gauss-Markov Theorem for the OLS estimators.
2. Suppose that the population mean and variance of employees' working hours per week in Singapore (SG) are μ_1 and σ_1^2 , respectively, and N_1 i.i.d. sample observations were drawn from this population; the population mean and variance of employees' working hours per week in Hong Kong (HK) are μ_2 and σ_2^2 , respectively, and N_2 i.i.d. sample observations were drawn from this population.

The SG and HK random samples are totally independent. We are interested in $\mu_1 - \mu_2$, the difference between the two population means, and use $\hat{d} = \bar{Y}_1 - \bar{Y}_2$ as its estimator, where $\bar{Y}_1$ is the sample average of the N_1 SG observations and $\bar{Y}_2$ is the sample average of the N_2 HK observations.

***The formulas of means, variances, and covariances of sums of random variables can be found in Key Concept 2.3 (p.74) of "Introduction to Econometrics".**

- (a) Compute $E[\bar{Y}_1]$, $E[\bar{Y}_2]$, $Var[\bar{Y}_1]$, and $Var[\bar{Y}_2]$.
- (b) Compute $E[\hat{d}]$, $Cov[\bar{Y}_1, \bar{Y}_2]$, and $Var[\hat{d}]$.
3. In this question, we show the reason why R^2 cannot be larger than 1.
 - (a) Show that $\sum_{i=1}^n \hat{u}_i X_i = 0$ by using the FOC of the OLS estimators.
 - (b) Show that $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$. [Hint: $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \sum_{i=1}^n \hat{u}_i (\hat{\beta}_0 + \hat{\beta}_1 X_i)$.]
 - (c) Show that $TSS = ESS + SSR$, where TSS and ESS are defined in the lecture slides, while $SSR = \sum_{i=1}^n \hat{u}_i^2$.
 - (d) Discuss why the OLS procedure is designed to have $\sum_{i=1}^n \hat{u}_i = 0$ and $\sum_{i=1}^n \hat{u}_i X_i = 0$. [Hint: There is something in the population that the OLS procedure would like to mimic by using $\sum_{i=1}^n \hat{u}_i = 0$ and $\sum_{i=1}^n \hat{u}_i X_i = 0$. Also, for two random variables X and Y , the covariance between X and Y can be written as:

$$\begin{aligned}
 Cov(X, Y) &= E[(X - \mu_x)(Y - \mu_y)] = E[XY - X\mu_y - \mu_x Y + \mu_x \mu_y] \\
 &= E[XY] - E[X]\mu_y - \mu_x E[Y] + \mu_x \mu_y = E[XY] - \mu_x \mu_y - \mu_x \mu_y + \mu_x \mu_y \\
 &= E[XY] - \mu_x \mu_y,
 \end{aligned}$$

where $\mu_x = E[X]$ and $\mu_y = E[Y]$.

4. This question follows the example of law school rankings discussed in the class. The dataset *lawsch_ranking.dta* contains the average starting salaries of students in each US law school, ranking of these law schools (No.1-142), average college GPA of the students in each school, etc.

Now we consider the following model:

$$salary_i = \beta_0 + \beta_1 top10_i + \beta_2 r11_25_i + \beta_3 r26_50_i + \beta_4 GPA_i + u_i, \quad (1)$$

where $i = 1, \dots, n$. Define the dummy variables $top10, r11_25, r26_50$ to be equal to 1 when the ranking of the i -th school falls into the appropriate range, and 0 otherwise.

- (a) Write down precisely the definition of each dummy variable.
- (b) Write down the linear regression model for the schools ranked Top 10, 11-25, 26-50, and 51-142, respectively.
- (c) Explain the meaning of the parameters $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ by using a graph.